

Intelligent Candidate Discovery & Ranking

Team: AI Pioneers

Objective:

- Rank 100,000 candidates for a Senior AI Engineer role
- Find true fit, ignoring keyword stuffers and honeypots
- Execute within a strict 5-minute CPU constraint

Architecture Overview

Given the 5-minute compute constraint on CPU, we opted for a highly optimized, single-pass feature extraction and heuristic scoring pipeline.

- Streaming JSONL reader to maintain a tiny memory footprint (<50MB).
- Custom Scoring Formula combining hard skills, experience, and behavior.
- Min-Heap data structure to maintain the top 100 candidates in $O(N \log K)$ time.
- Zero external network calls or LLM API dependencies during ranking.

Feature Engineering & Scoring

We score candidates based on the actual requirements in the JD:

- Title Relevance: Regex matching for AI/ML/Software Engineer.
 - Severe penalties for irrelevant titles (e.g., Marketing, HR).
- Experience Fit: Targets 5-9 years. Rewards product company experience.
 - Penalizes careers spent entirely at consulting firms.
- Skill Match: Weights Vector DBs, Eval frameworks (NDCG), and Python.
 - Multiplier based on proficiency and duration used.

Behavioral Signals & Honeypots

The platform signals are critical to finding a 'hirable' candidate.

- Ghost Penalties: Heavily penalize low recruiter response rates.
- Inactivity Penalties: Penalize candidates who haven't logged in >180 days.
- Engagement Boosts: Reward high GitHub activity scores.

Honeypot Detection (Traps):

- Automatically flags candidates with multiple 'expert' skills but 0 months used.
- Disqualifies candidates with massive mismatches between titles and AI skills.

Results & Output

- Runtime: Evaluates all 100,000 candidates in under 10 seconds.
- Reasoning Generation: Dynamically generates a specific, fact-based reasoning string for each top 100 candidate without hallucinating.

Example Reasoning:

'Senior AI Engineer with 6.5 yrs exp. Has product exp. Skills: pinecone, python. Resp rate: 0.90. Active GitHub (75.0).'

Fully reproducible and aligned with the hackathon rules.